Molecular Dynamics using Scipion-Chem

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Introduction

Dynamic **simulations of the biomolecules** of interest mimicking **real conditions**: solvated, ions, temperature, pressure…



atomic positions r(t)

↓

$F = -\nabla U(r)$

↓

integrate F = ma

↓

atomic positions r(t+Δt)

↓

trajectory r(t), r(t+Δt), r(t+2Δt)...

Introduction

The force is the negative gradient of the potential energy *U*:

$$F_i = -\nabla_i U(r_1, \ldots, r_n)$$

$$U = U_{\text{bonded}} + U_{\text{non-bonded}}$$

Calculated using forcefields. Collection of empirical function - spring constants, partial charges, atomic radii - fitted to quantum calculations and experiment.

BONDED forces



Bonds

springs



Angles

springs



Dihedrals

periodic

NON-BONDED forces



van der Waals

Lennard-Jones



Electrostatics

Coulomb

KNOWN · INPUT

Number of atoms ***N***

Masses of all atoms m_j

Initial positions r_j: {*x*, *y*, *z*}

Forcefield

PER ATOM · EVERY STEP

velocity v_j

acceleration a_j

new position r'_j

energy U_j force $f_j = -\nabla U_j$

WHOLE SYSTEM · EVERY STEP

kinetic, potential & total energy

thermodynamic variables P, V, T

Short steps of **~2 fs**, repeated millions of times. ⟶ **Thousands/millions of steps builds the trajectory r(t)**

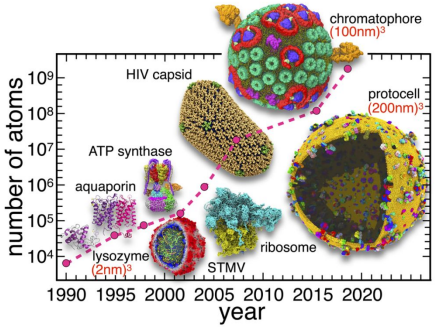
Computationally expensive

Simulation time
Time step Δt ≈ 2 fs
small enough to resolve the fastest motions (bond vibrations).

System Size

Accessible timescales *fs → ms*

fs	ps	ns	µs–ms
bond vibration	side chains	loops	domains, folding



Computational cost scale with the number of atoms and the number of steps
solvated protein ≈ 30–100k atoms -> 100 ns ≈ 50M steps

WARNING: MD results are statistical - run several replicas before drawing conclusions.

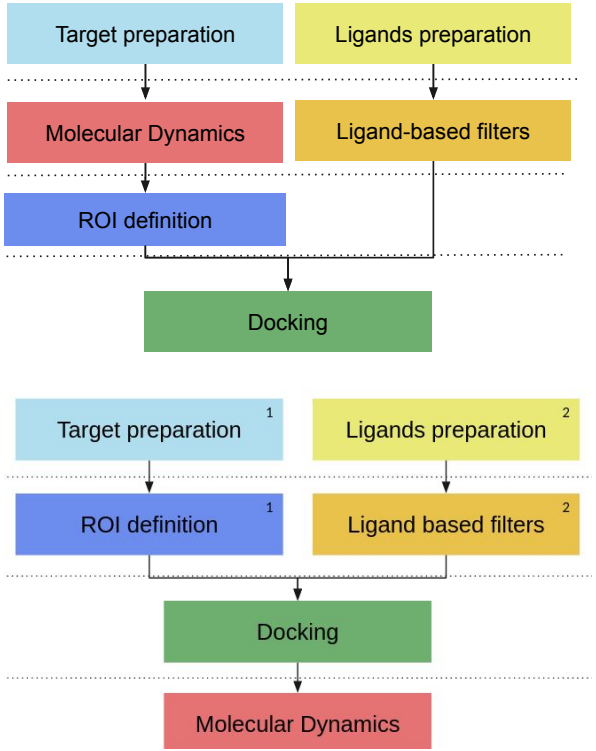
Molecular dynamics for VDS

1) Dynamics of **receptor**

- Simulating the receptor for checking stability, flexibility, cryptic pockets…
- Using different conformations as docking target (ensemble docking)

2) Dynamics of **complex**

- Simulating the **complex** to study their behavior together.
- Check **RMSD**, **contacts** during trajectory - is the pose stable?
- Estimate the **interacting energies**



The molecular dynamics workflow

1 Prepare structure
Clean the PDB: fix missing atoms/loops, add hydrogens, set protonation (pH), keep/curate the ligand.

2 Parametrise
Assign protein force field; generate ligand parameters (GAFF/CGenFF) + partial charges (AM1-BCC/RESP).

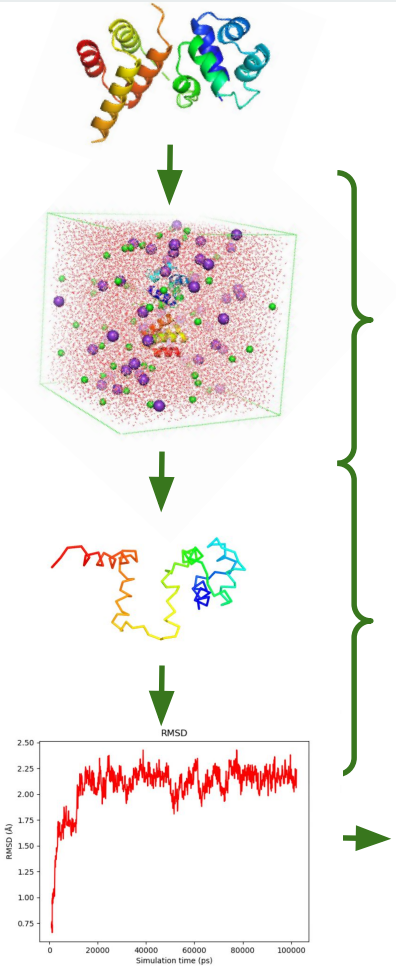
3 Solvate & neutralise
Place molecule in a periodic box, add explicit water (TIP3P/SPC), add ions for charge & ionic strength.

4 Minimise
Steepest-descent energy minimisation removes clashes and bad contacts.

5 Equilibrate
NVT (heat to desired T) then NPT (set density/pressure); relax solvent around the solute; prevent energy explosions

6 Production
Remove restraints; run the long unbiased simulation.

7 Analysis
Extract useful information from the trajectory



Chem

System Preparation protocol

MD simulation protocol

Modification protocol

Viewers

Analysis protocols

Three Engines - One Plugin for each



FAST. FLEXIBLE. FREE.
GROMACS

OPEN SOURCE

Extremely fast C/C++/SIMD engine. Dominant for proteins. CPU + GPU.

Topology	.top / .itp
Coordinates	.gro
Run input	.mdp → .tpr
Trajectory	.xtc / .trr
Usage	command line

AMBER

ACADEMIC / COMMERCIAL

Most common FF, famous for ligand binding analysis. Strong GPU.

Topology	.prmtop
Coordinates	.inpcrd / rst7
Run input	mdin
Trajectory	.nc (NetCDF)
Usage	command line



OPEN SOURCE

Python/C++ library & toolkit. Highly flexible, scriptable, custom forces. Gaining importance

Topology	.pdb
Coordinates	.pdb / .cif
Run input	Python script
Trajectory	.dcd / .xtc
Usage	Python API

Plugin differences

- Rich restraints definitions
- Fastest
- MMPBSA/MMGBSA calculations

- Automatic membrane setup
- Support of user defined input running files

- Constant pH simulations
- Several ligand parametrization options
- OpenFE FEP calculations

Interoperability

- Format conversions
- Import Molecular Dynamics System
- Analysis protocols

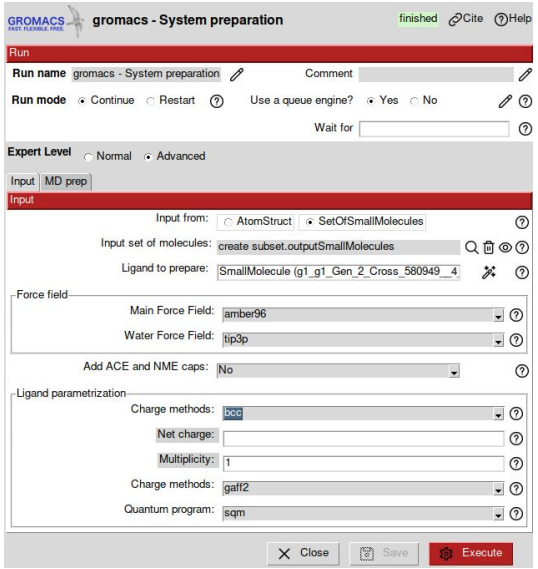
Example GROMACS - System preparation protocol

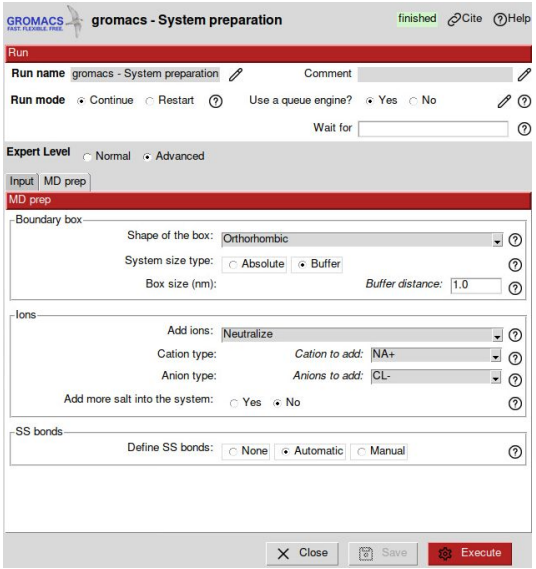
One protocol turns a raw structure into a simulation-ready system.

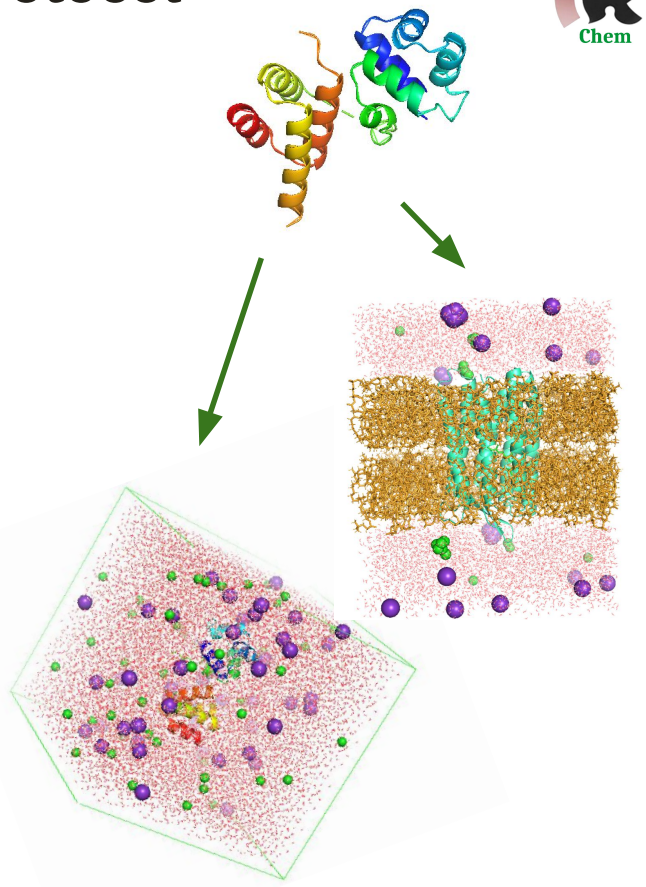
Input can be a **docked molecule** or an **atomic structure**.

The options are **endless** - current parameters are:

force field	main + water (amber96, tip3p)	**ions**	neutralise or concentration
ligand	charges (bcc), GAFF2, QM (sqm)	**SS bonds**	automatic or manual detection
box	shape & size	**Adding caps**	to gaps



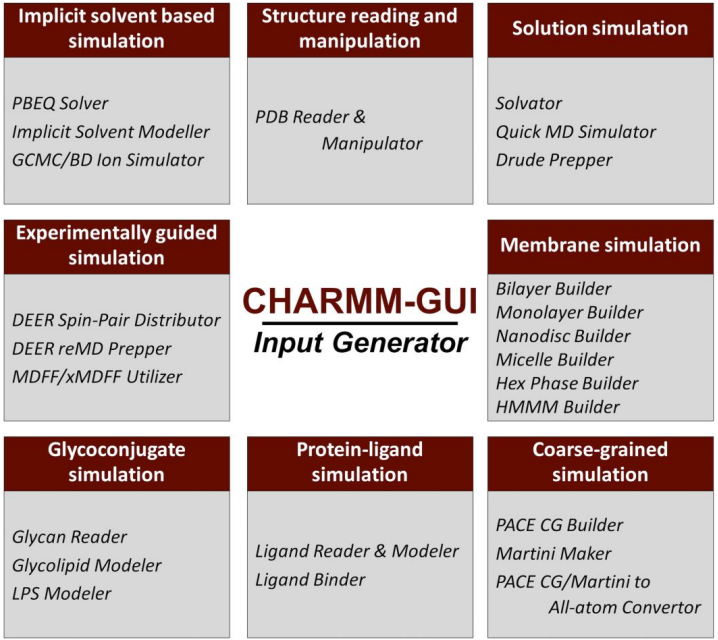




Example GROMACS - Import MD system

Improve **Import MDSystem** protocol

CHARMM-GUI useful for complex system generation



- Structure
- Topology
- Index
- Parameterized ligands

Run and analyze in scipion

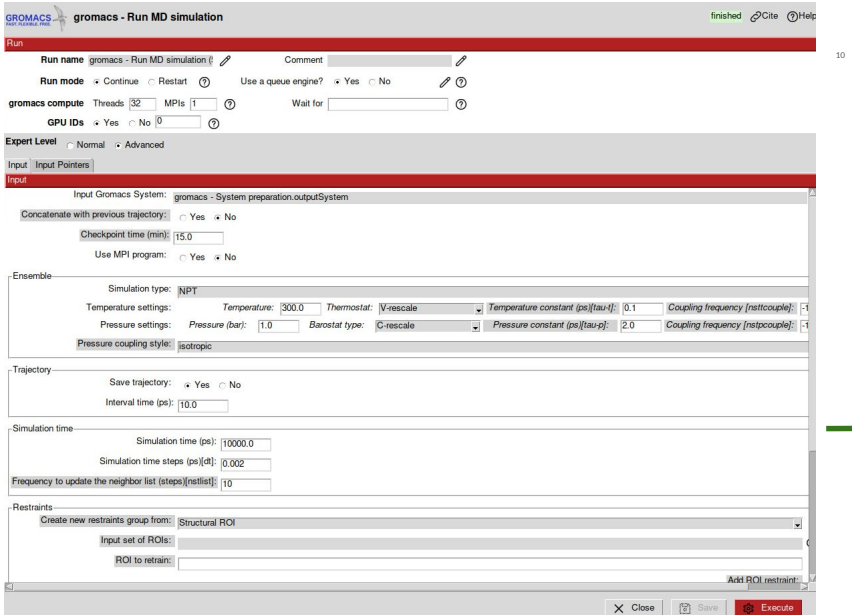
Widely used and very complete system preparation tool

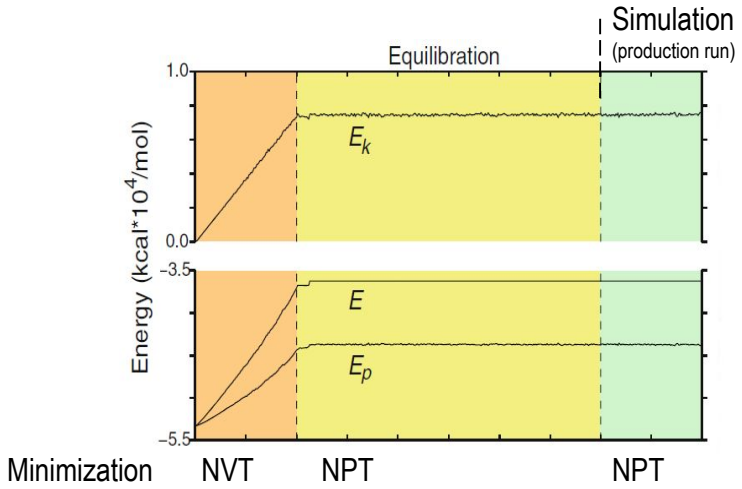
Example GROMACS - Run MD simulation

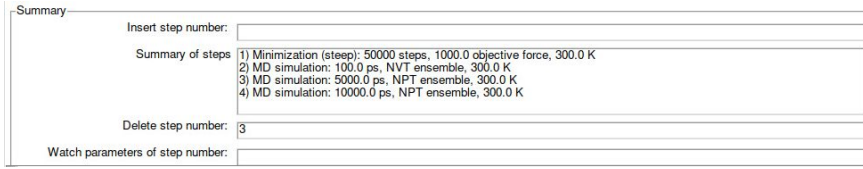
Minimisation
Steepest descent — relax clashes before any dynamics.

NVT
Fixed atoms, volume & temperature — heat the system to T.

NPT
Fixed atoms, pressure & temperature — settle density before production.







Example GROMACS - Modify MDSystem

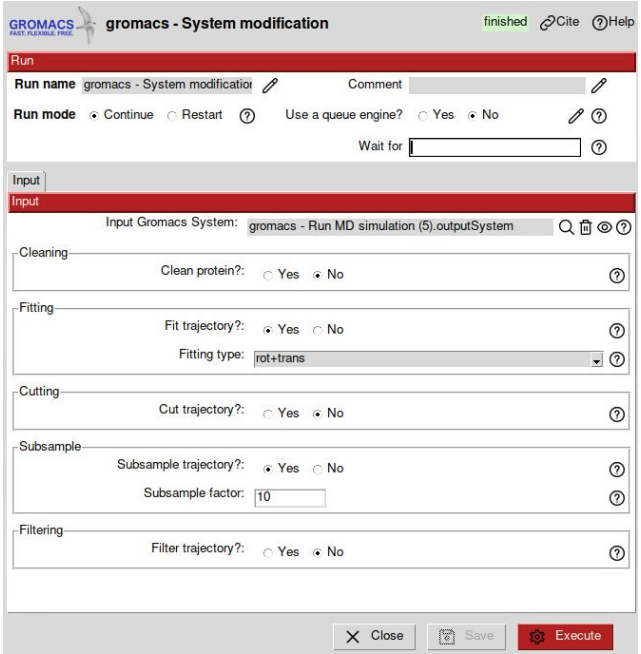
A toolbox to clean up a raw trajectory before analysis.

Fit
remove rotation + translation

Cut
trim to a frame window

Subsample
keep every N-th frame

Clean & filter
strip solvent, smooth noise

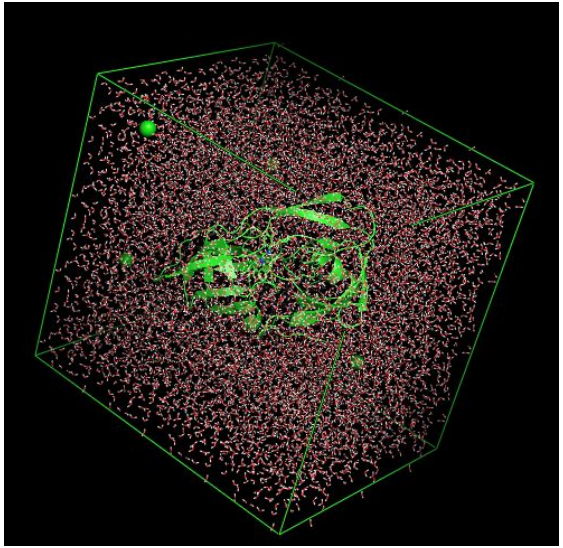


Trajectory analysis

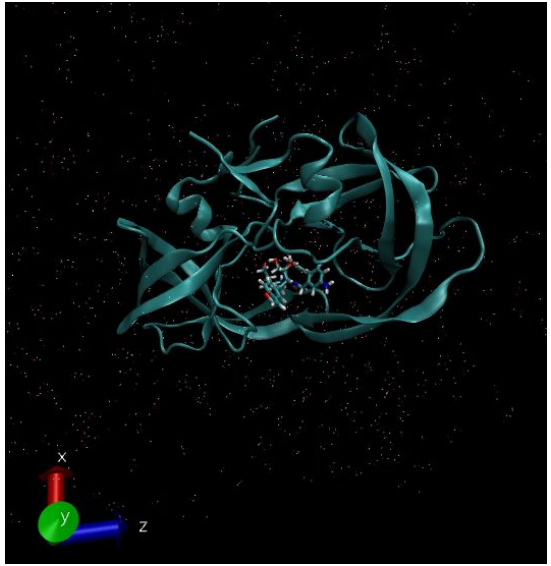
Mainly done in the viewer + some analysis protocols.

Visually inspect the system and trajectory in **PyMOL** and **VMD**.

PyMOL - view system



VMD - view trajectory

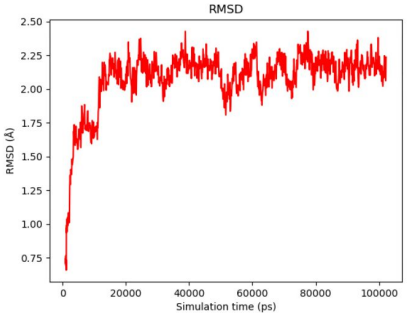


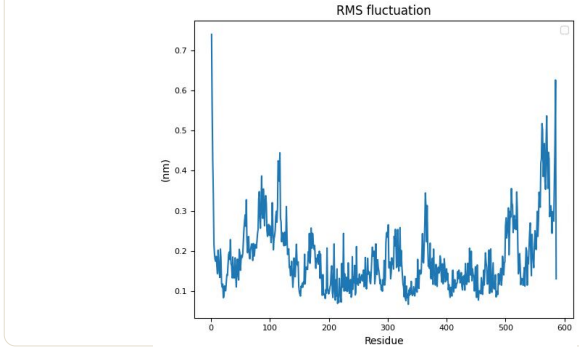
Movement analysis - RMSD & RMSF

RMSD — first check for equilibrabation

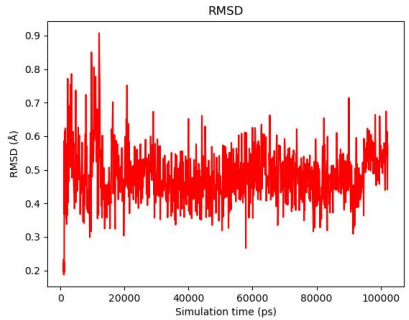
RMSF — movement per atom

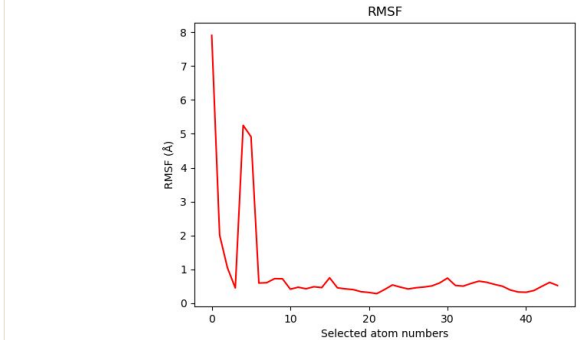
PROTEIN





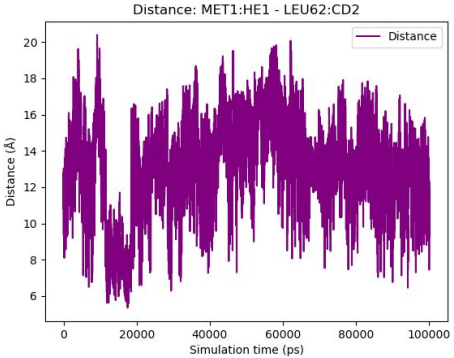
LIGAND



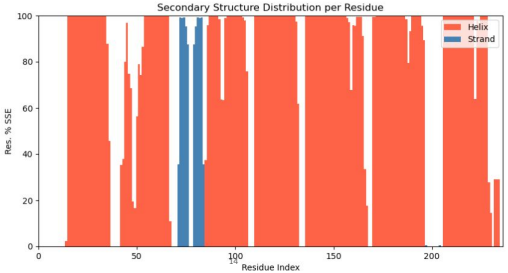


Protein movement analysis

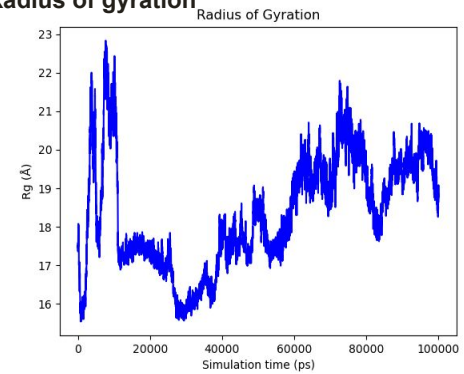
Atom distances



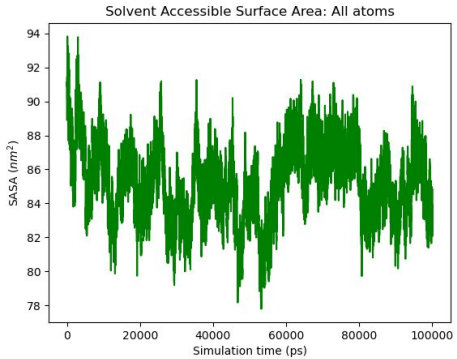
Secondary structure



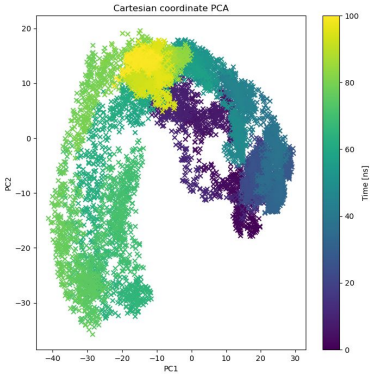
Radius of gyration



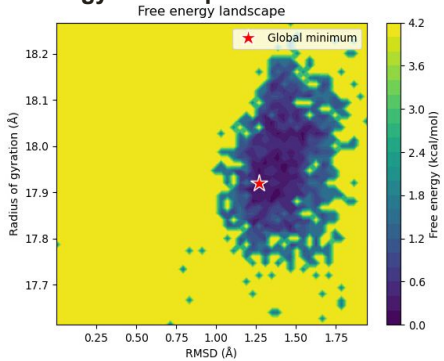
SASA



PCA



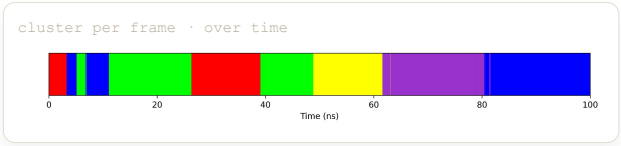
Free-energy landscape

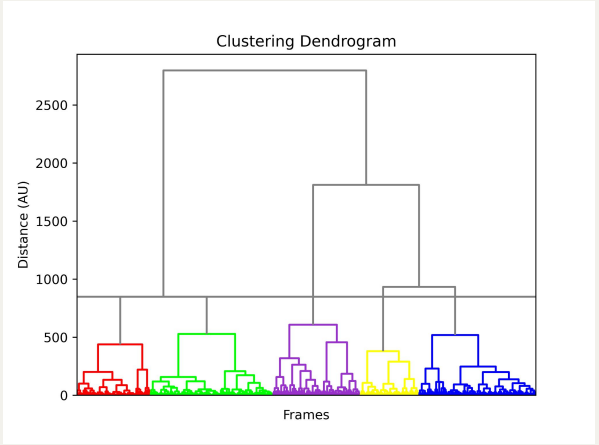


Protein movement analysis - Trajectory clustering protocol



Extract representative structures

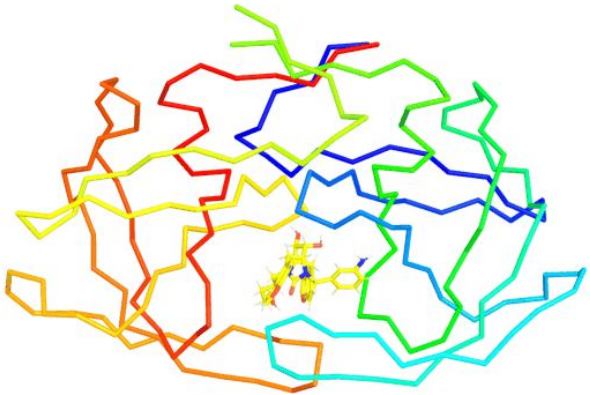




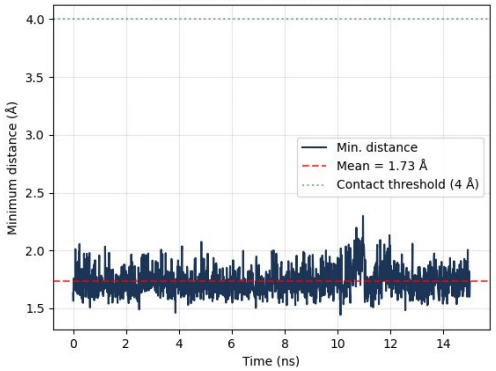
Useful for **ensemble docking**:

- Docking towards a set of conformations of a target
- Accounts for its flexibility

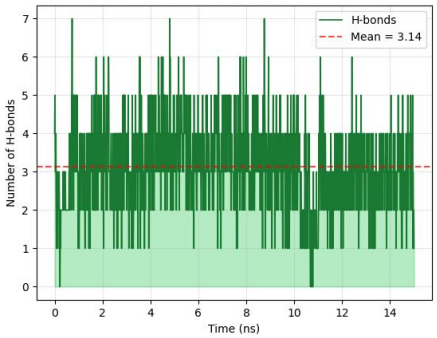
Protein-Ligand binding: Pose stability



Protein-Ligand minimum distance



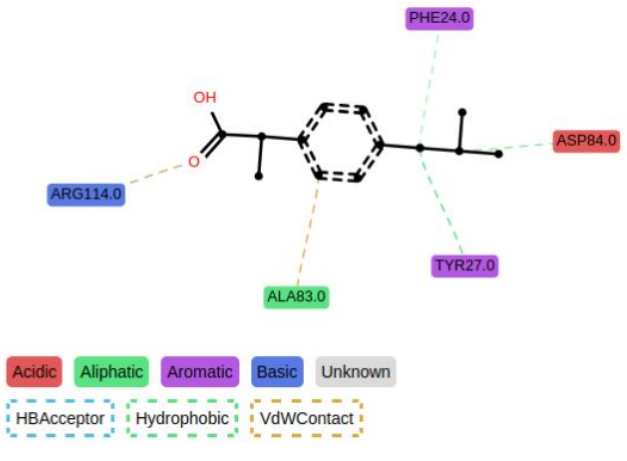
Protein-Ligand H-bonds



Protein–ligand binding analysis: interaction fingerprints OpenMMDL and ProLIF protocol

Every interaction type - hydrophobic, van der Waals, H-bond, π-stacking - mapped for each interacting residue for each frame.





Binding energy calculations

SECONDS

Docking scores

Fast scoring of static poses

HOURS

MM/GBSA

End-point method over the MD trajectory, generalised Born solvation

GROMACS

HOURS

MM/PBSA

End-point method over the MD trajectory, Poisson Boltzmann solvation

GROMACS

DAYS ON GPUs

FEP

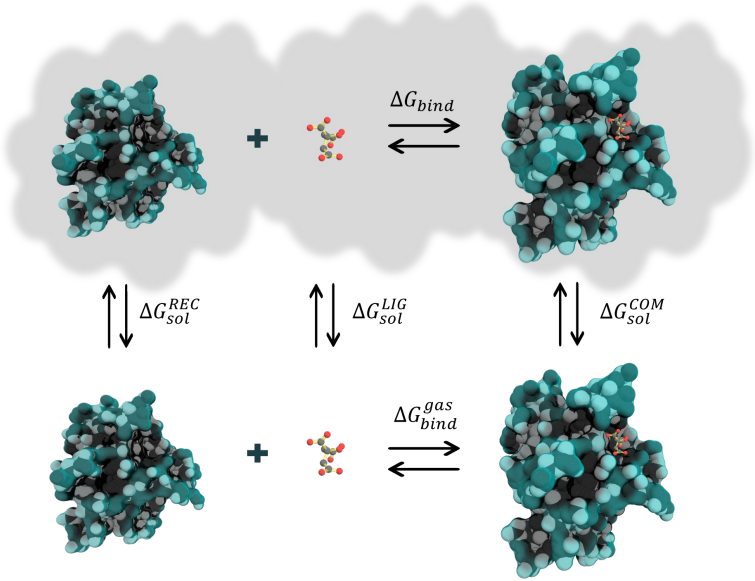
Alchemical binding free energies, state-of-the-art accuracy

OpenMM

Faster, cheaper → More accurate

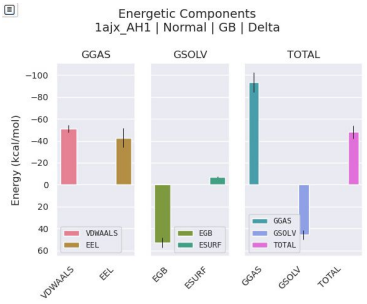
Binding energy calculations - MMPBSA & MMGBSA

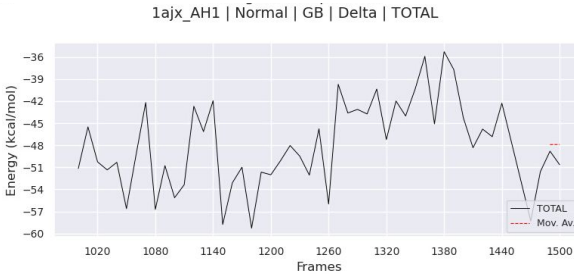
ΔG_bind = ΔG_bind(vacuum) + ΔG_solv_COM − ΔG_solv_REC − ΔG_solv_LIG



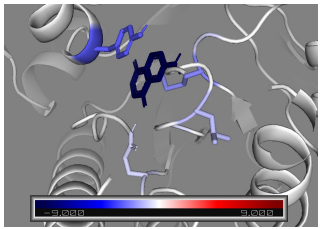
Valdés-Tresanco et al., 2021, *Journal of chemical theory and computation*

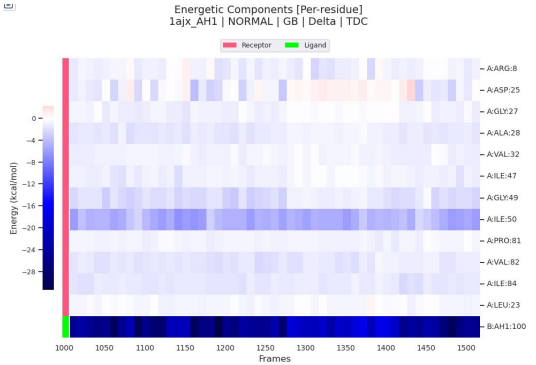
Energy contributions and energy evolution



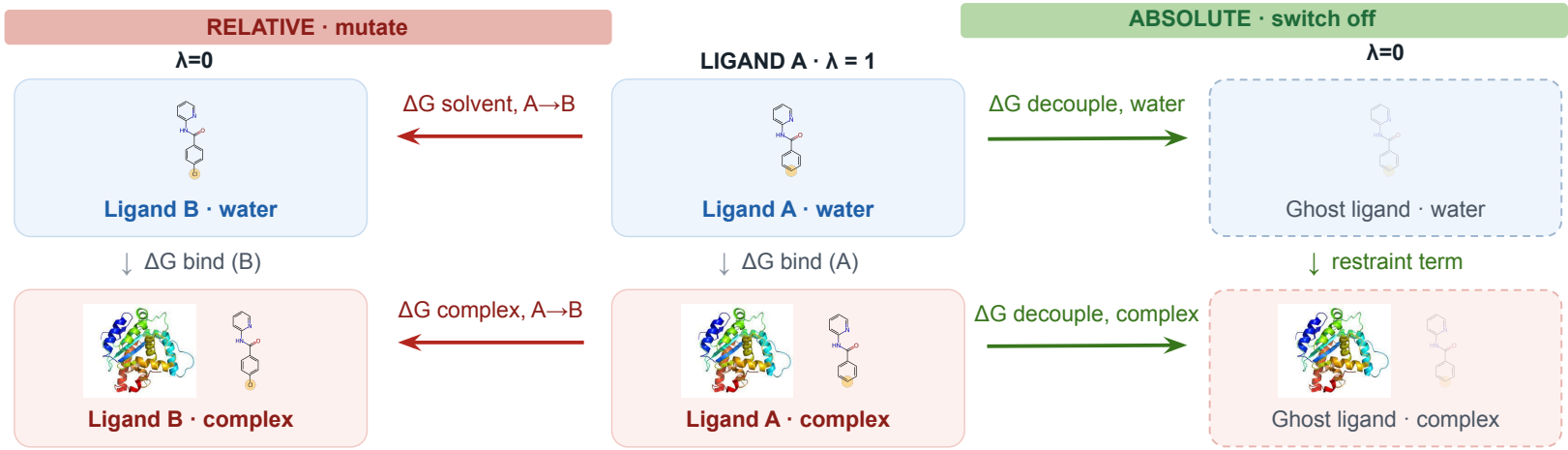


Residue decomposition





Binding energy calculations - FEP



ΔΔG bind = ΔG complex − ΔG solvent

Ranks close analogues in lead optimisation

ΔG bind = ΔG dec, water − ΔG dec, complex + restraints

Absolute affinity of one ligand · costlier, needs restraints



ligand	dG (kcal/mol)	uncertainty (kcal/mol)
E31_902	-0.99	0.1
E42_901	-0.44	0.08
E43_903	1.21	0.09
E44_904	2.7	0.1
J23_900	-2.5	0.2

4 days 18 hours

E31_902: dG_bind = -9.53 +/- 0.81 kcal/mol

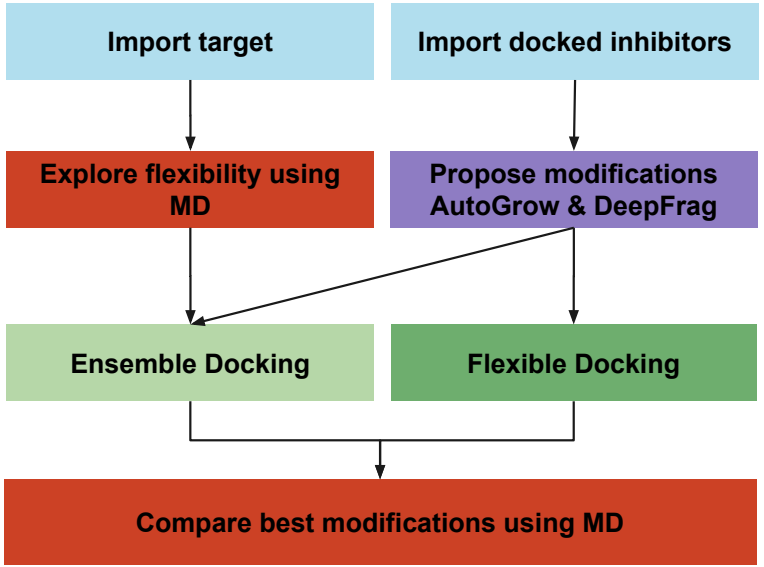
4 days 2 hours

HANDS-ON: Lead optimization of PI3Kα inhibitors

Project: **FineTuningPI3Ka_course**

Objective: propose molecules from the the PI3Kα with increased affinity

1. Extract clusters from the target MD only simulation
2. Ensemble docking
3. Run MD simulations of PI3Kα-Top hits and the original inhibitors & compare



Proposed exercises

1. Run a Molecular dynamic simulation of another inhibitor of PI3Kα:
    a. Duplicate the whole MD simulation block buutt select a different molecule as input
    b. Make the Run MD simulation protocol run a shorter simulation. Play with all the options.
    c. Hint: If some System preparation fails try running *convert structure format* protocol using Docked ligands option and PDB output format
2. Compare the Ligand-Target contacts of this new inhibitor to the ones obtained in Serabelisib simulation

Periodic Boundry conditions

